

TRACE: IF A IS A SQUARE MATRIX, THEN THE TRACE OF A IS DENOTED BY $\text{tr}(A)$ AND IS DEFINED TO BE THE SUM OF THE ENTRIES ON THE MAIN DIAGONAL OF A .

EXAMPLE:

FOR $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \rightarrow \textcircled{1}$

$$\text{tr}(A) = 4 + 1 + 1 = 6$$

LAST TIME WE SAW THAT EIGENVALUES OF $A = 1, 2, 3$
 $\lambda_1 \quad \lambda_2 \quad \lambda_3$

CONSIDER

$$\lambda_1 + \lambda_2 + \lambda_3 = 1 + 2 + 3 = 6 = \text{tr}(A)$$

RESULT: IF A IS A SQUARE MATRIX THEN $\text{tr}(A) = \text{SUM}$ OF ITS EIGENVALUES.

ALSO NOTE THAT FROM $\textcircled{1}$

$$\det(A) = 4 + 1(2) = 6 = \lambda_1 \lambda_2 \lambda_3$$

2) RESULT: IF A IS A SQUARE MATRIX THEN $\det(A) =$ PRODUCT OF ITS EIGENVALUES.

TRY THE FOLLOWING:

(a) SHOW THAT THE CHARACTERISTIC EQUATION OF A 2x2 MATRIX A CAN BE EXPRESSED AS. $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0 \rightarrow \textcircled{1}$

(b) IF $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ THEN THE SOLUTIONS OF THE CHARACTERISTIC EQUATION OF A ARE

$$\lambda = \frac{1}{2} \left[(a+d) \pm \sqrt{(a-d)^2 + 4bc} \right]$$

NOTE: IF λ_1, λ_2 ARE ROOTS OF $\textcircled{1}$ THEN

$$\lambda_1 + \lambda_2 = \text{tr}(A), \lambda_1 \lambda_2 = \det(A)$$

RECALL: FOR α, β AS ROOTS OF $ax^2 + bx + c = 0, a \neq 0$
 $\alpha + \beta = -\frac{b}{a}$ AND $\alpha\beta = \frac{c}{a}$

3) DIAGONALIZATION P. 347
P. 365 (7th ED.) (6th ED.)
→ P. 347

DEFINITION: A SQUARE MATRIX A IS CALLED DIAGONALIZABLE IF THERE IS AN INVERTIBLE MATRIX P SUCH THAT $P^{-1}AP$ IS A DIAGONAL MATRIX; THE MATRIX P IS SAID TO DIAGONALIZE A .

EXAMPLE:

$$= \begin{bmatrix} 2 & 0 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ -1 & -2 & 1 \\ -1 & -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 2 & 0 \\ -3 & -4 & 1 \\ -3 & -4 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

P IS THE MATRIX HAVING EIGENVECTORS OF A AS ITS COLUMN VECTORS AND $P^{-1}AP$ IS A DIAGONAL MATRIX HAVING EIGENVALUES ON THE MAIN DIAGONAL.

Application of DIAGONALIZATION (4)

THE EIGENVECTOR PROBLEM.

GIVEN AN $n \times n$ MATRIX A , DOES THERE EXIST A BASIS FOR $\mathbb{R}^n$ CONSISTING OF EIGENVECTORS OF A ?

THEOREM 7.2.1

P. 347 $\rightarrow$ 6TH ED.

P. 365 $\rightarrow$ 7TH ED.

IF A IS AN $n \times n$ MATRIX, THEN THE FOLLOWING ARE EQUIVALENT.

- (a) A IS DIAGONALIZABLE
- (b) A HAS n LINEARLY INDEPENDENT EIGENVECTORS.

PROOF: (b) $\Rightarrow$ (a)

ASSUME THAT A HAS n LINEARLY INDEPENDENT EIGENVECTORS $\underline{p}_1, \underline{p}_2, \dots, \underline{p}_n$, WITH CORRESPONDING EIGENVALUES $\lambda_1, \lambda_2, \dots, \lambda_n$.

CONSIDER THE MATRIX P WITH $\underline{p}_1, \underline{p}_2, \dots, \underline{p}_n$ AS ITS COLUMN VECTORS.

i.e.

(5)

$$P = \begin{bmatrix} \overset{\underline{P_1}}{\downarrow} P_{11} & \overset{\underline{P_2}}{\downarrow} P_{12} & \dots & \overset{\underline{P_n}}{\downarrow} P_{1n} \\ P_{21} & P_{22} & \dots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \dots & P_{nn} \end{bmatrix}$$

NOW THE COLUMNS OF THE PRODUCT AP ARE $\underline{AP_1}, \underline{AP_2}, \dots, \underline{AP_n}$

EXTRA: CONSIDER $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

AND $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$, $AP = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$

$$\Rightarrow AP = \begin{bmatrix} a_{11}P_{11} + a_{12}P_{21} & a_{11}P_{12} + a_{12}P_{22} \\ a_{21}P_{11} + a_{22}P_{21} & a_{21}P_{12} + a_{22}P_{22} \end{bmatrix}$$

$\swarrow \quad \quad \quad \searrow$
 $\hookrightarrow P_{11} \quad \quad \quad \hookrightarrow P_{21}$

FIRST COLUMN OF AP IS $\begin{bmatrix} a_{11}P_{11} + a_{12}P_{21} \\ a_{21}P_{11} + a_{22}P_{21} \end{bmatrix}$

$$= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} P_{11} \\ P_{21} \end{bmatrix} = \underline{AP_1}$$

BUT $A\underline{P}_1 = \lambda_1 \underline{P}_1$, $A\underline{P}_2 = \lambda_2 \underline{P}_2, \dots$,
 $A\underline{P}_n = \lambda_n \underline{P}_n$, SO THAT

$$AP = \begin{bmatrix} \lambda_1 P_{11} & \lambda_2 P_{12} & \dots & \lambda_n P_{1n} \\ \lambda_1 P_{21} & \lambda_2 P_{22} & \dots & \lambda_n P_{2n} \\ \vdots & \vdots & & \vdots \\ \lambda_1 P_{n1} & \lambda_2 P_{n2} & \dots & \lambda_n P_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1n} \\ P_{21} & P_{22} & \dots & P_{2n} \\ \vdots & \vdots & & \vdots \\ P_{n1} & P_{n2} & \dots & P_{nn} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

$$= \underset{\substack{\uparrow \\ P}}{PD} \underset{\substack{\uparrow \\ D}}{\quad}$$

$\Rightarrow \therefore \boxed{AP = PD}$, WHERE D

IS THE DIAGONAL MATRIX HAVING
 THE EIGENVALUES $\boxed{\lambda_1, \lambda_2, \dots, \lambda_n}$
 ON THE MAIN DIAGONAL. SINCE THE
COLUMN VECTORS OF $\boxed{P}$ ARE LINEARLY
INDEPENDENT, THEREFORE

$\boxed{\text{RANK}(P) = n}$ SO THAT

$\boxed{\det(P) \neq 0}$, 

RECALL THAT RANK IS ALSO
DEFINED AS THE HIGHEST ORDER
OF THE NONZERO DETERMINANT.

$\therefore P$ IS INVERTIBLE;

THUS $AP = PD$ CAN BE WRITT-
EN AS $P^{-1}AP = D$; THAT IS,
 A IS DIAGONALIZABLE.

CONVERSE IS ALSO TRUE.

$(a) \Rightarrow (b)$

IF A IS DIAGONALIZABLE
THEN A HAS n LINEARLY
INDEPENDENT EIGENVECTORS

AND THEY FORM A
BASIS FOR R^n